The Welfare Effects of Providing Information about Externalities

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Motivation

Externalities → market failures from undersupply of prosocial behaviors

- Taxes restore first-best, but face constraints to adoption

Common alternative: **provide information** about the externality

1. **Product labels** certify socially responsible production
2. **Mass-media campaigns** for: vaccination, pollution reduction, etc.
3. **Demand reduction programs** promote resource conservation

Hundreds of studies evaluate impacts of information on behavior
(Schultz et al. 2007; Jensen 2010; Allcott 2011; Dupas 2011; Ferraro & Price 2013; Wiswall & Zafar 2014; Beshears et al. 2015; Perez-Truglia & Troiano 2018; Bursztyn et al. 2019; Alsan & Eichmeyer 2021; see Haaland et al. 2023 for a review)

- but little is known about **welfare**

How a "Protect others" flyer affects welfare



- Δ Consumer welfare $> 0 \Rightarrow$ "win-win"
- Δ Consumer welfare $\ll 0 \Rightarrow \uparrow$ vaccination but $\downarrow$ welfare

This paper

What are the welfare effects of providing information about externalities?

Context: carbon emissions

Theory: adapt canonical prosocial signaling model (Bénabou & Tirole 2006)

- Provides testable predictions and a framework for quantifying welfare

Experiments: design interventions using estimates of participants' carbon footprint

- Elicit WTP for each intervention $\rightarrow$ consumer welfare impact
- Estimate abatement responses to each intervention $\rightarrow$ external damage reduction
- Welfare effects relative to (i) no policy and (ii) an abatement-equivalent tax

Three information interventions

Impact-based info

Conveys the external impacts of one's behavior

Social comparison info

Compares one's behavior to others

Public recognition

Discloses one's behavior to others



Three information interventions

Impact-based info

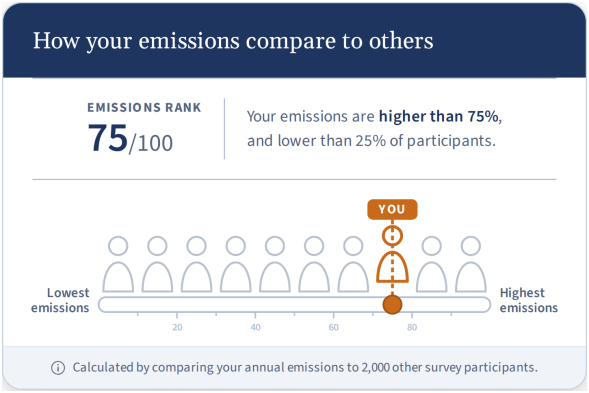
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Three information interventions

Impact-based info

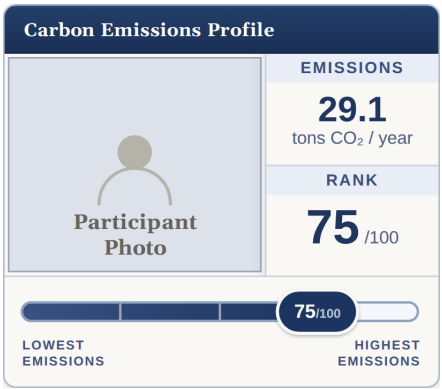
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Preview of results

1. **Participants value private information**
   - Positive WTP for both impact-based and social comparison information
2. **Preferences for public recognition depend on relative emissions**
   - WTP positive for low emitters, negative for high emitters
3. **Public recognition reduces emissions**
   - not statistically significant for private information
4. **Behavior change can be a poor proxy for welfare**
   - despite increased abatement, public recognition reduces total welfare

Contributions

1. Impact of information on environmental behaviors Allcott (2011); Ferraro & Price (2013); Allcott & Rogers (2014); Jessoe & Rapson (2014); Benartzi et al. (2017); Ito et al. (2018); Gillingham & Stock (2018); Dechezleprêtre et al. (2025)

→ Combine behavioral response with consumer impacts to estimate welfare

2. Welfare effects of nonfinancial policy instruments DellaVigna, List & Malmendier (2012); Allcott & Taubinsky (2015); Damgaard & Gravert (2018); Bernheim & Taubinsky (2018); Allcott & Kessler (2019); Thunström (2019); Butera et al. (2022); Karing (2024); Allcott et al. (2025)

→ **impact-based information**

→ social comparison information on a **pure public good**

→ public recognition of a **negative externality**

3. Preferences for information and information avoidance Loewenstein (1987); Kőszegi (2006); Dana, Weber & Kuang (2007); Karlsson et al. (2009); Oster et al. (2013); Golman et al. (2017); Grossman & van der Weele (2017)

→ Measure WTP for information about a large real-world externality

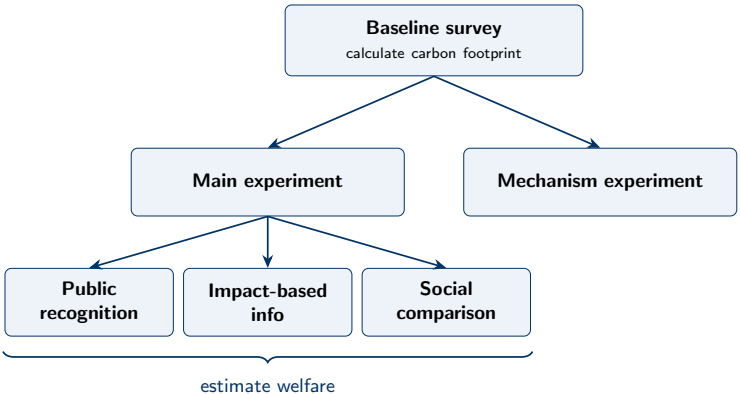
Roadmap

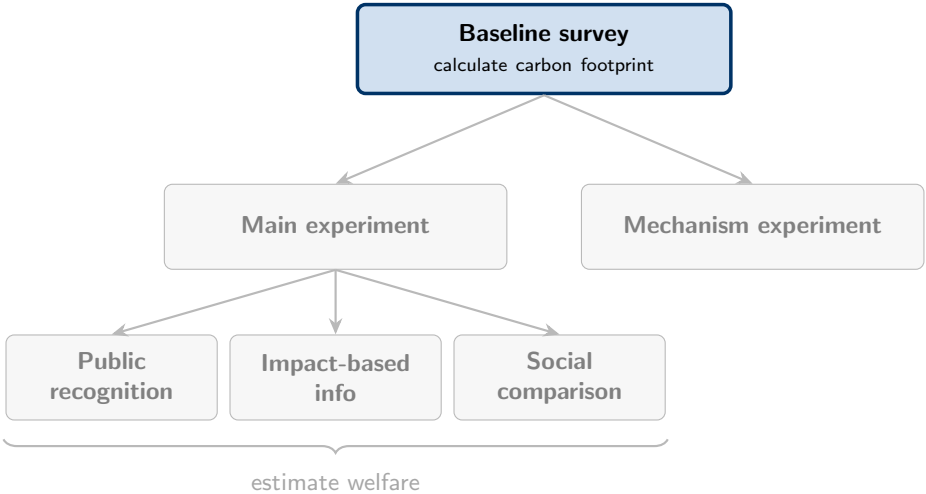
1. **Experimental design**
2. **Model**
3. **Preferences for information**
4. **Abatement effects**
5. **Welfare**

Experimental Design

Sample and overview of experiments

Sample: ∼6,500 U.S. adults recruited from online survey platform Prolific



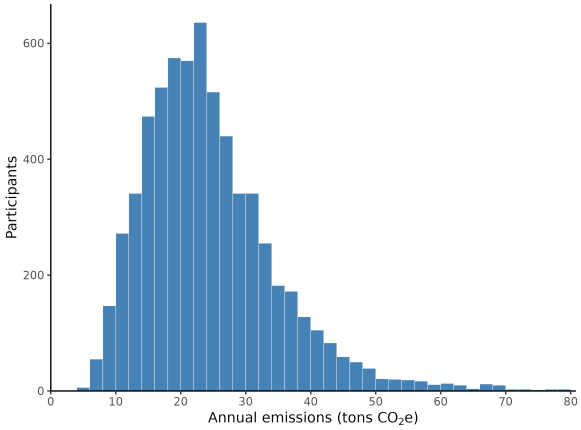

Baseline survey
calculate carbon footprint
Main experiment
Mechanism experiment
Public recognition
Impact-based info
Social comparison
estimate welfare

Baseline survey: Calculating participants' carbon footprint

∼3 min survey

- Location
- Income
- Transportation
- Energy use
- Diet

Emissions calculator



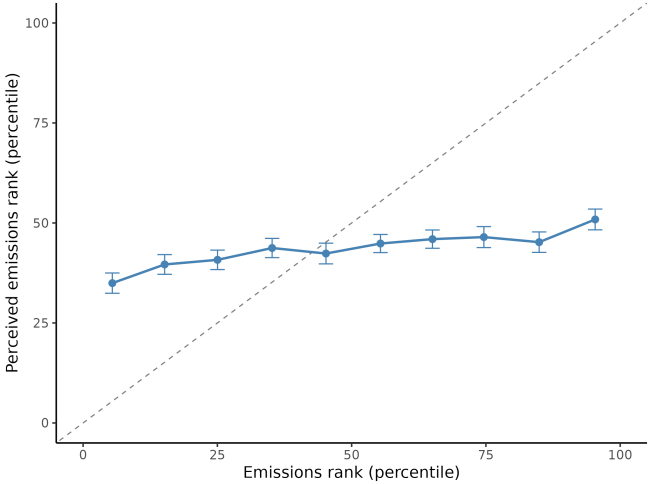
Emissions rank:
percentile in participant distribution

Annual damages:
emissions × \$185/ton SCC

(Rennert et al. 2022)

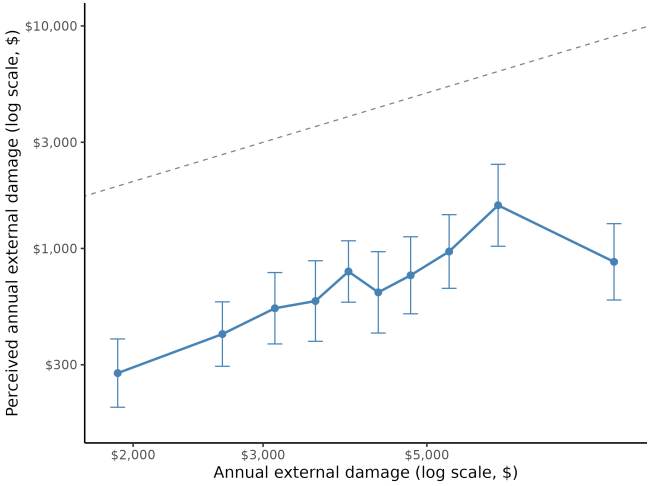
How informed are participants
about relative emissions and annual damages?

Low emitters overestimate relative emissions, high emitters underestimate

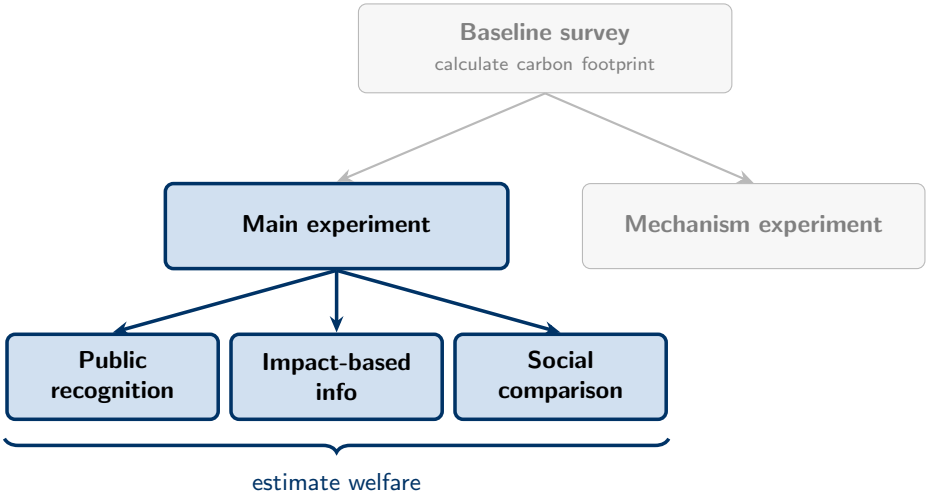


- Lowest decile emitters: overestimate by 30 percentiles
- Highest decile emitters: underestimate by 44 percentiles

Participants underestimate annual external damages



- 83% of participants underestimate annual damages
- Actual damages are ∼6x perceived damages on average


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Main experiment

Separate arm for each intervention

WTP elicitation

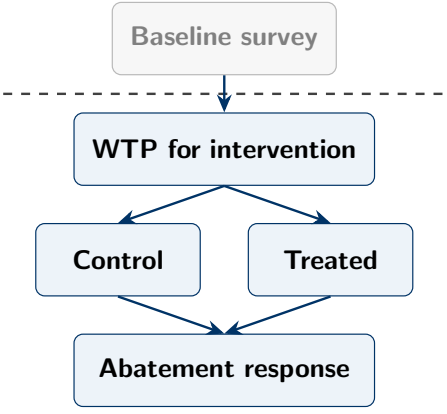
- WTP to receive or avoid the intervention

Treatment assignment

- randomly assigned (90% of participants)
- assigned via BDM (10% of participants)

Abatement response

- public recognition: structurally estimated
- private information: measured directly through emissions offset contribution exercise



Revealed preference measure of consumer welfare impacts

Challenge: consumer welfare impacts may depend on the information's realization

- e.g., positive for low emitters, negative for high emitters

Strategy method: elicit WTP conditional on possible realizations (Butera et al. 2022)

- *Information scenarios*: possible realizations of the information
  - Public recognition and social comparison: possible emissions ranks
  - Impact info: possible $ damages
- WTP conditional on actual realization affects outcome (10% of participants)

Elicitation example: Public recognition

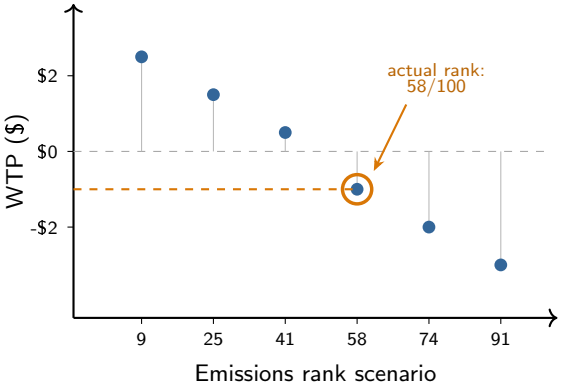
If your emissions rank is **X/100**, *would you prefer your emissions profile to be shown to another participant or kept anonymous?*

- ○ *Show my profile to another participant*
- ○ *Keep my profile anonymous*

For six possible emissions ranks (X/100)

- elicit binary preference
- WTP for preferred outcome

⇒ WTP at the actual rank measures the **consumer welfare impact**



Model

Model setup

Emissions abatement: choose abatement $a_i \in [0, e]$ given baseline emissions e

Motives:

- Abatement costs $c(a; \eta_i)$: $c'(a) = \eta_i + k\,a$
- Altruism θ_i: share of external damages internalized
- Image utility: desire to be seen, by others and self, as a high-θ type

Signaling:

- Type (θ_i, η_i) known only to the *decision-making self*
- External observers and *evaluative self* see a_i then infer θ_i
- High abatement may reflect high altruism or low abatement costs

The abatement choice problem

Individual i chooses a_i to maximize:

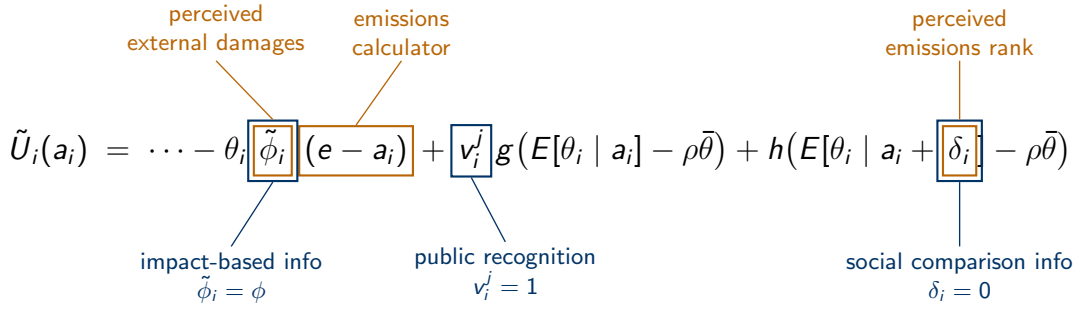
$$\tilde{U}_i(a_i) = \underbrace{y_i}_{\text{income}} - \underbrace{c(a_i;\, \eta_i)}_{\text{cost}} - \underbrace{\theta_i\, \tilde{\phi}_i\, (e - a_i)}_{\text{altruism}} + \underbrace{v_i^j\, g\big(E[\theta_i \mid a_i] - \rho\bar{\theta}\big)}_{\text{social-image}} + \underbrace{h\Big(\tilde{E}[\theta_i \mid a_i] - \rho\bar{\theta}\Big)}_{\text{self-image}}$$

- $\tilde{\phi}_i$: perceived marginal external damage
- $v_i^j \in \{0, 1\}$: a_i revealed to observer j
- g', $h' > 0$; $\quad g(0) = h(0) = 0$
- $E[\theta_i \mid a]$: inferred altruism type
- $\rho\bar{\theta}$: standard for positive image utility
- $\tilde{E}[\theta_i \mid a_i] = E[\theta_i \mid a_i + \delta_i]$
- δ_i: inaccurate self-inference
  - $\delta_i > 0$: misperception of relative emissions

Mapping model to experiments

Measured

Interventions



Information directly impacts consumer welfare through inferred type

Normally distributed types and quadratic image utility $\Rightarrow$ equilibrium inference mapping:

$$E[\theta_i \mid a_i] = \bar{\theta} + D(\tilde{\phi})\left(a_i - \bar{a}(\tilde{\phi})\right)$$

- Diagnosticity $D(\tilde{\phi}) \geq 0$
- Benchmark abatement $\bar{a}(\tilde{\phi})$

Diagnosticity is increasing in perceived marginal damages ($D'(\tilde{\phi}) > 0$)

- larger perceived damages $\Rightarrow$ amplified altruism incentives (relative to cost shocks)

Testable model predictions

Public recognition

- **Prediction 1:** WTP^{PR} is decreasing in emissions, positive at low and negative at high emissions

Social comparison info

- **Prediction 2:** WTP^{SC} is decreasing in emissions rank, positive for lower than anticipated emissions ranks, and negative for higher than anticipated emissions ranks

Impact-based info: WTP^{IB} is the sum of:

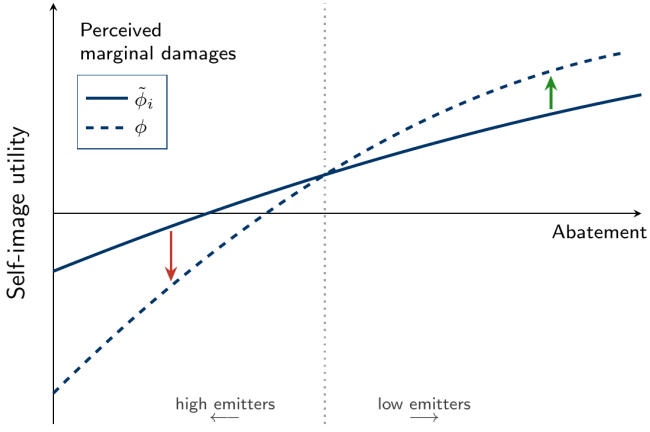
1. Instrumental value $(+)$: increasing in the change in abatement
2. Self-image effects $(+/-)$: decreasing in external damages for high emitters (increasing for low emitters)

Impact-based info directly affects image utility

Increasing perceived marginal damages $\rightarrow$

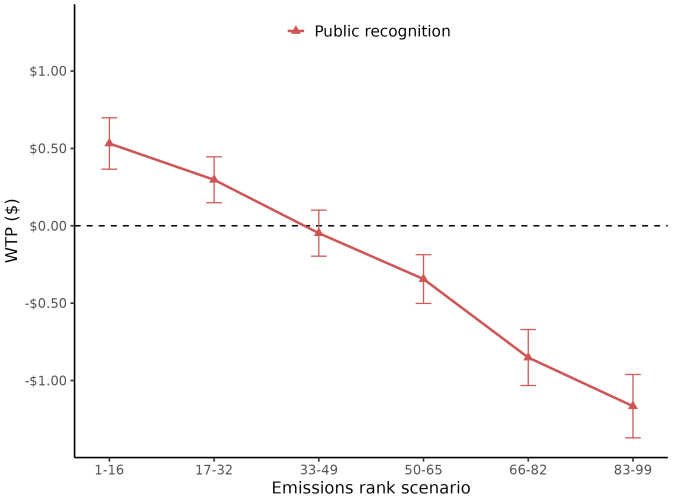
- shifts right benchmark $\bar{a}(\tilde{\phi})$
- increases diagnosticity $D(\tilde{\phi})$

$\Rightarrow \downarrow$ **image utility** for high emitters and $\uparrow$ **image utility** for low emitters



Preferences for Information

WTP for public recognition is decreasing in emissions rank

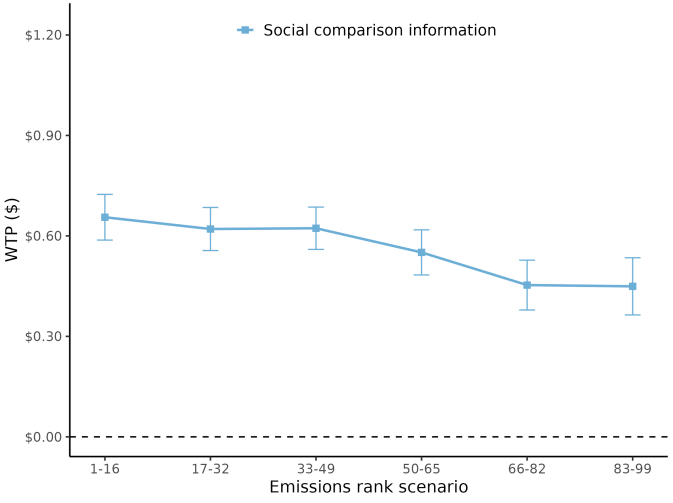


Prediction 1: WTP^{PR} decreasing in rank, positive at low and negative at high

Result: consistent

- positive in bottom tercile
- negative above median

WTP for social comparison info is decreasing in emissions rank

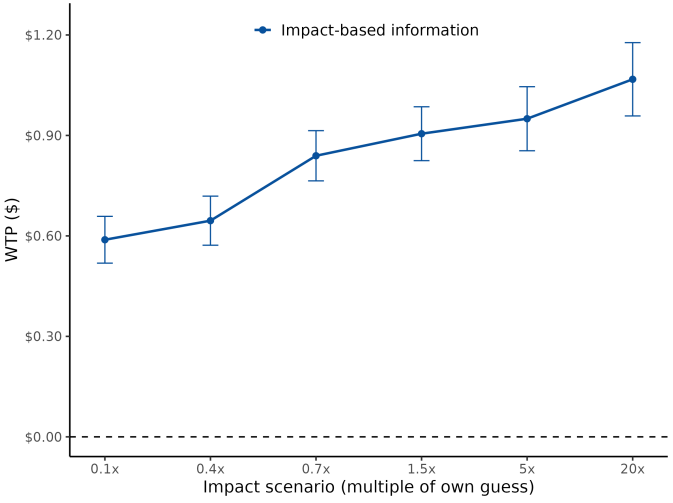


Prediction 2: WTP^{SC} decreasing in rank, negative at higher than anticipated ranks

Result: partially consistent

- decreasing in rank
- but positive at all ranks

WTP for impact-based info is increasing in impact scenario

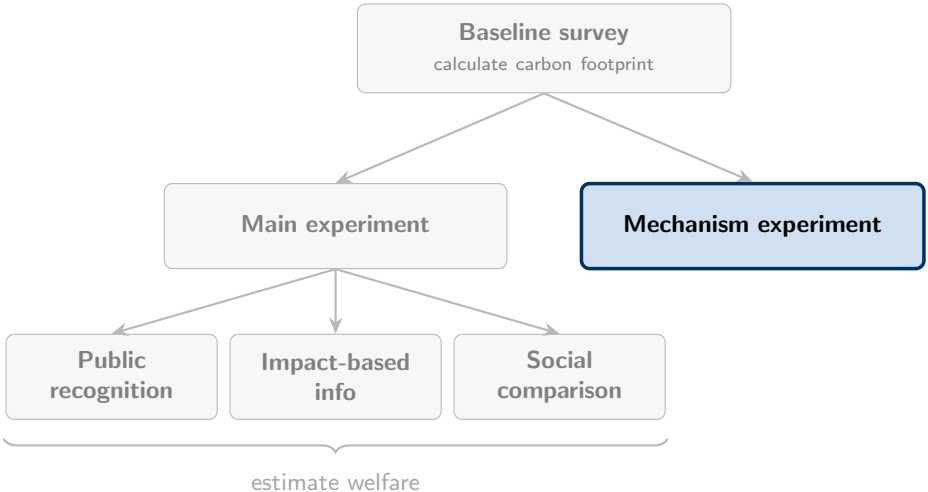


Self-image effects?

- No (similar WTP slope between low and high emitters)

Instrumental value?

- Potentially (if behavior change is increasing in impact scenario)

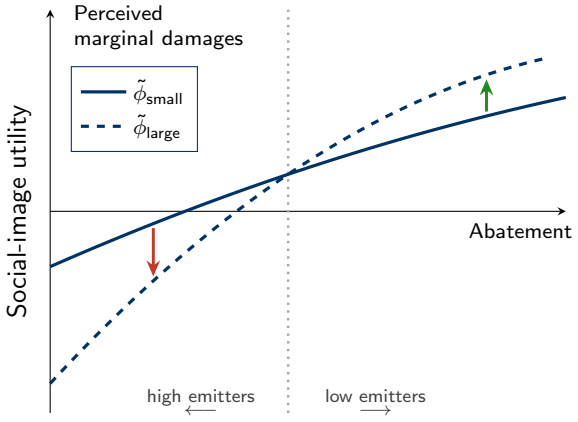

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Do marginal damage perceptions affect social image utility?

Two observers with perceived marginal damages $\tilde{\phi}_{\text{small}} < \tilde{\phi}_{\text{large}}$

- $\tilde{\phi}_{\text{small}}$: abatement weak signal of altruism type
- $\tilde{\phi}_{\text{large}}$: abatement strong signal of altruism type

Prediction 3: Low (high) emitters prefer observers who perceive marginal damages as large (small)



Mechanism experiment

Each **subject's** emissions profile is shown to one **observer**

Observers differ in their stated views on damages

- *"Which of the following best describes your personal views on carbon emissions? Damages from carbon emissions are. . . "*

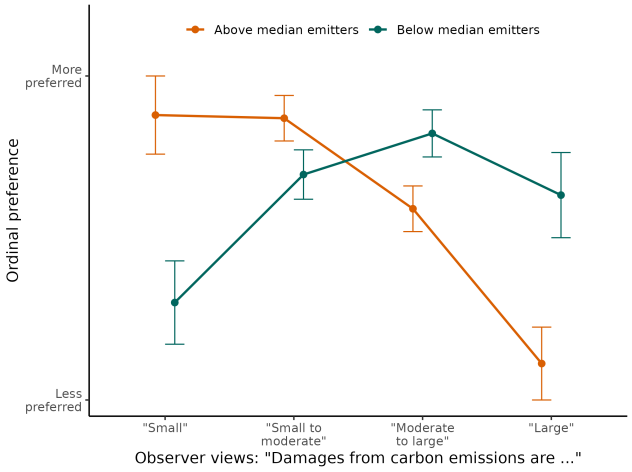
Subjects see their own emissions profile, then decide

- ordinal preference ranking of possible observers
- WTP for more preferred observer
  - multiple price list on observer pairs

Choices affect which observer sees profile



High emitters prefer observers who think damages are small

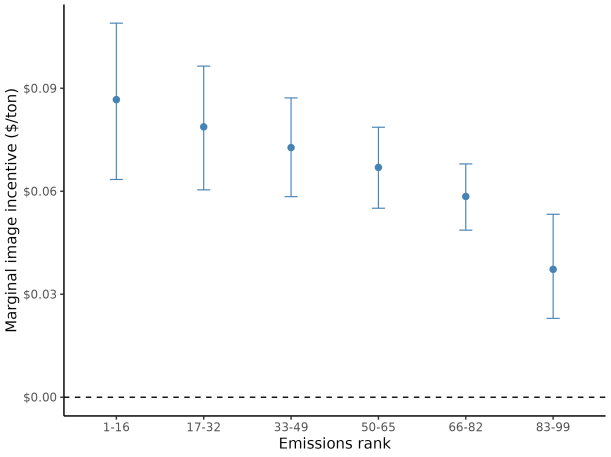


Consistent with Prediction 3,

- **High emitters** prefer observers who believe damages are **smaller**
- **Low emitters** prefer observers who believe damages are **larger**

Abatement Effects

Public recognition creates an incentive to abate



Marginal image incentive ($\partial g / \partial a$)

- slope of WTP^{PR} (implicit subsidy)
- **small** ($\approx$ \$0.07/ton), **but positive** $\Rightarrow$ public recognition increases abatement

Measuring private information's effect on abatement

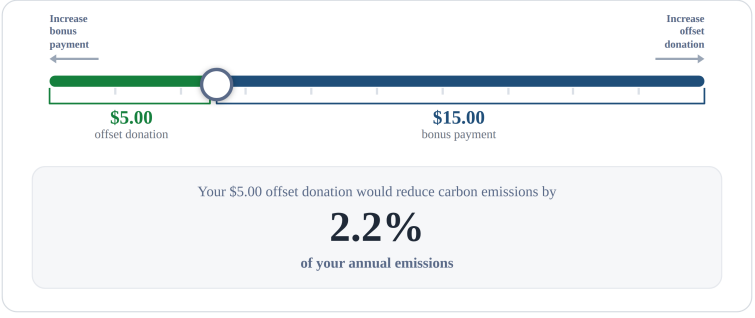
Voluntary contribution exercise:

Divide $20 between (i) additional bonus payment and (ii) emissions offset contributions

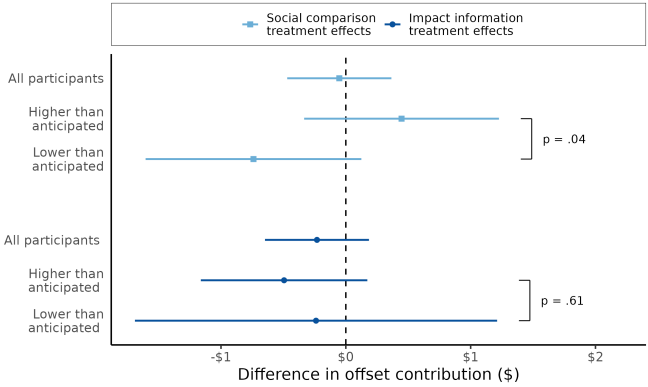
- implemented for 5% of participants

Contributions used to retire regulated emissions permits

- subsidized price of $10/ton



Private information does not increase abatement



- Aggregate effects are not statistically significant
- Offsetting effects from learning emissions rank is higher vs lower than anticipated

Welfare

Welfare effects relative to two counterfactuals

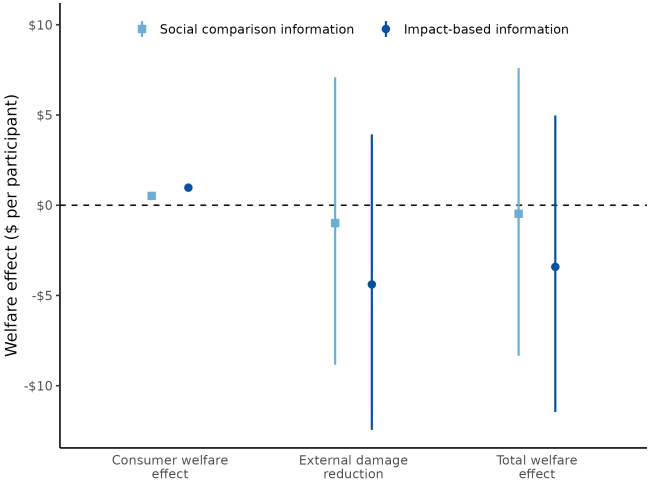
vs. no intervention: total welfare effect = consumer welfare effect + external damage reduction

- consumer welfare effect: WTP under actual information scenario
- external damage reduction: abatement response × \$185/ton SCC

vs. abatement-equivalent tax (lump-sum redistribution):

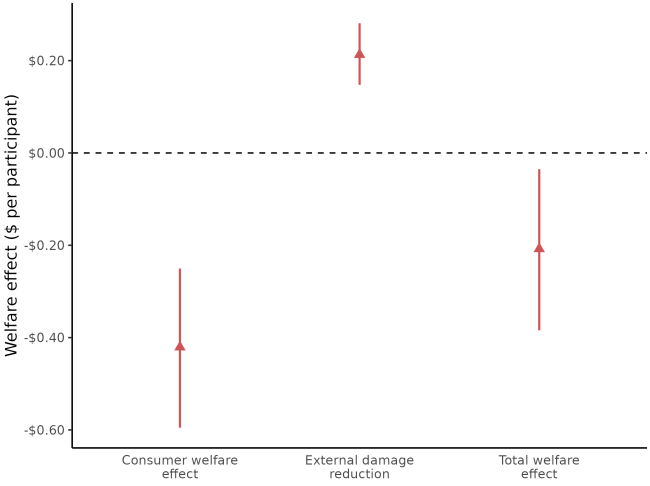
- abatement equivalent ⇒ damage reduction nets out
- direct effects on consumer welfare remain

Welfare effects of private information relative to no intervention



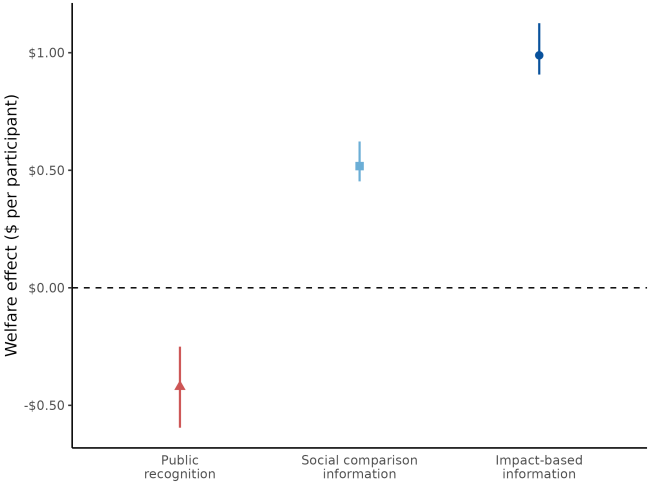
- Consumer welfare increases
- External damage effects are relatively large and uncertain
- Total welfare effects not statistically significant

Welfare effects of public recognition relative to no intervention



- Reduces consumer welfare
- Increases abatement (reduces external damages)
- Total welfare effect **negative**

Welfare effects relative to abatement equivalent tax



- Public recognition **underperforms** an abatement-equivalent tax
- Private information **outperforms** an abatement-equivalent tax

Conclusion

Information affects consumer welfare

- Private information: increases consumer welfare, larger gains from impact-based information
- Public recognition: utility gains for low emitters, losses for high emitters, negative-sum in aggregate

Behavior change is not a sufficient statistic for welfare effects

- Private information does not increase abatement, but increases consumer welfare
- Public recognition increases abatement, but reduces total welfare

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